

From the Theory of Cognition to the Theory of Knowledge Architecture

Why the Future of Artificial Intelligence May Depend Less on Larger Models and More on Better Foundations of Knowledge

Stefan Bruns, July 2026

Introduction

The previous whitepapers in this series explored memory, experience, meaning, significance, pattern formation, and semantic gravitation.

Together, they pursued a central question:

How can artificial systems develop stable cognitive structures over time?

The investigation began with memory.

It continued through identity and continuity to meaning and significance.

Most recently, it examined how recurring patterns may condense into stable semantic structures.

At first glance, these concepts appear to be building blocks of a theory of artificial cognition.

Yet as the discussion deepens, a different connection begins to emerge.

The true value of these concepts does not lie in replacing probability.

It lies in improving the quality of the knowledge foundations upon which probabilities are calculated in the first place.

This is where the importance of **Knowledge Architecture** begins.

1. The Wrong Starting Question

Many discussions about artificial intelligence focus on outcomes.

Can a model generate better answers?

Can it solve more complex problems?

Can it outperform human experts?

The common response is often to pursue more performance and larger storage capacities.

These questions always direct our attention toward the end of a process.

Yet if we seek more reliable and more stable answers, we must first improve the quality of the foundations from which those answers emerge.

Let us stay within the context of scientific research for a moment, although the principle can be applied far beyond it.

Research does not begin with answers.

Research begins with observations.

With experiments.

With measurements.

With comparisons.

All of these elements can be transformed into reproducible patterns.

And these patterns can then become part of future calculations and analyses.

The quality of an insight is therefore not determined first by its answer.

It is determined by the structure of the knowledge foundation from which that answer emerges.

The crucial question becomes:

How can a system organize knowledge and information in such a way that insight emerges—insight that ultimately leads to more reliable answers?

2. Probability Is Not the Problem

Modern AI systems are built upon probabilities.

They compare vast amounts of available information and calculate the outcomes that appear statistically most plausible.

This principle has enabled remarkable progress.

Language models.

Drug discovery.

Simulations of complex systems.

The limitation, however, does not lie in probability itself.

It lies in the quality of what is being compared.

A calculation can only be as reliable as the knowledge foundation on which it is based.

Progress therefore does not arise from larger models alone.

Progress emerges from better knowledge structures.

3. Why Research Requires Comparable Experiences

Consider the example introduced in Whitepaper 5.

Three researchers investigate the boiling point of water.

The first observes boiling at 100°C at sea level.

The second measures 93°C during a mountain expedition.

The third documents 85°C in the Himalayas.

Each observation is correct.

Each observation represents a memory of a specific event.

Yet none of these observations explains anything on its own.

Only when these observations are compared does insight emerge.

Comparison produces experience.

Multiple experiences produce meaning.

The resulting insight is simple:

Water does not always boil at the same temperature.

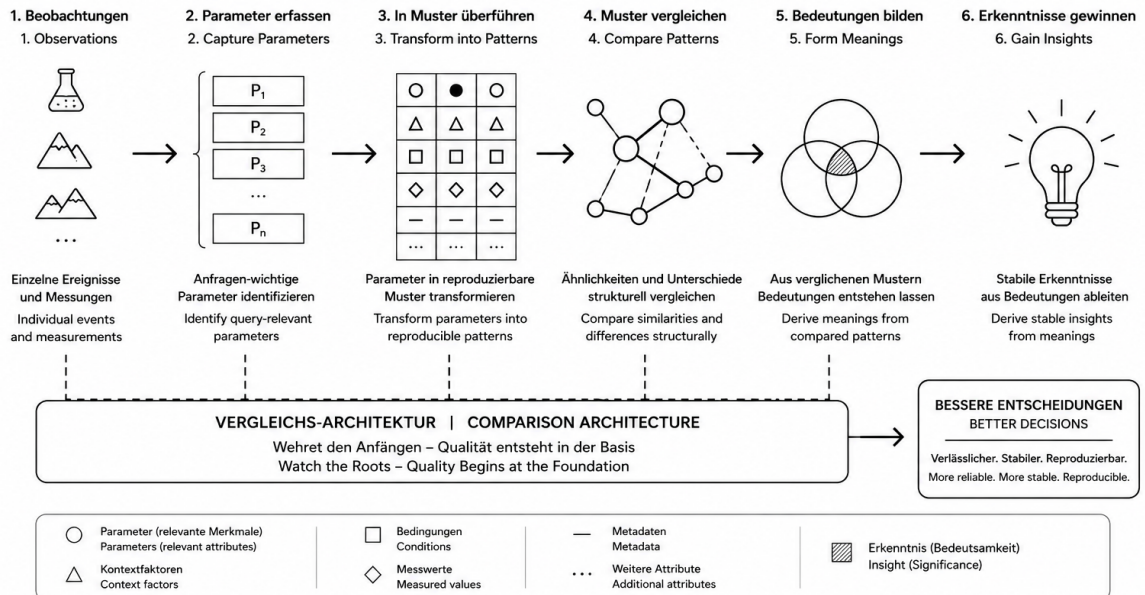
The discovery does not reside in the individual observation.

It resides in the comparison of observations.

Von der Beobachtung zur Erkenntnis

From Observation to Insight

Die Architektur des Vergleichs als Grundlage belastbarer Erkenntnisse
The Architecture of Comparison as the Foundation of Reliable Insights



from Observation to Insight

4. The Architecture of Comparison

This principle can be found in almost every scientific discipline.

Physics.

Chemistry.

Biology.

Materials science.

Pharmaceutical research.

Aerospace engineering.

A single observation rarely creates reliable knowledge.

Knowledge emerges when experiences can be compared with one another.

This introduces a new architectural requirement.

Experiences must become structurally comparable.

To achieve this, their underlying parameters must be transformed into reproducible patterns.

Without comparability, there can be no condensation.

Without condensation, there can be no meaning.

Without meaning, there can be no significance—and ultimately no insight.

When we create comparable patterns already at the level of memories and experiences, we improve the foundation from which future insights emerge.

This is what I refer to as **semantic condensation**—or, in a broader sense, **semantic gravitation**.

5. Semantic Condensation as Knowledge Infrastructure

Semantic condensation serves a different purpose than is often assumed.

Its primary function is not storage optimization.

Nor is it the reduction of data volumes.

Its true purpose is to enable comparability.

Experiences are transformed into patterns.

Patterns are weighted according to their qualitative relevance.

The comparison of similar patterns produces meanings.

The comparison of multiple meanings produces significance.

Only then do larger spaces of insight become visible.

Through condensation, reconstructable knowledge structures emerge that extend far beyond individual events.

6. Semantic Gravitation and the Quality of Insight

Semantic gravitation describes the process through which certain patterns gain relevance over time.

Within scientific systems, it serves an additional function.

It acts as a qualitative filter.

Not every observation carries the same epistemic value.

Not every experiment contributes equally to the understanding of a phenomenon.

Some findings are repeatedly confirmed.

Others remain isolated cases.

Some patterns appear across disciplines.

Others lose their significance under closer examination.

Semantic gravitation describes the stabilization of those patterns that repeatedly demonstrate explanatory power.

7. From Data Quality to Insight Quality

The development of artificial intelligence often focuses on:

- larger datasets
- larger models
- greater computational resources

These factors remain important.

Yet they address only part of the challenge.

The next stage of artificial systems will depend on how effectively knowledge can be organized.

The decisive factor is not the quantity of available information.

It is the quality of the relationships between pieces of information.

Not computational power alone.

But epistemic structure.

8. Knowledge Architectures

At this point, the distinction between cognition and knowledge architecture becomes visible.

The theory of cognition asks:

How does a system remember?

How does it form experiences?

How does meaning emerge?

The theory of knowledge architecture asks a different question:

How must these processes be organized in order for reliable insights to emerge?

At first glance, the difference appears small.

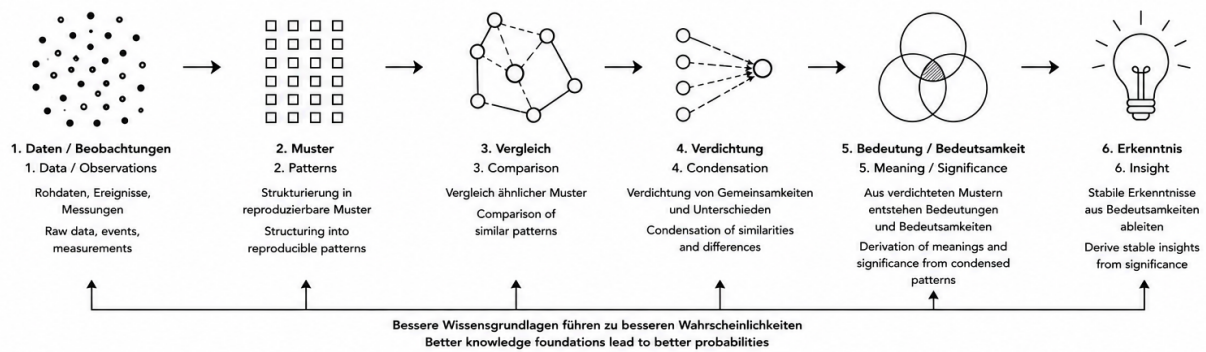
In reality, it marks a shift in perspective.

The focus is no longer the emergence of cognition itself.

The focus becomes the emergence of reproducible insight.

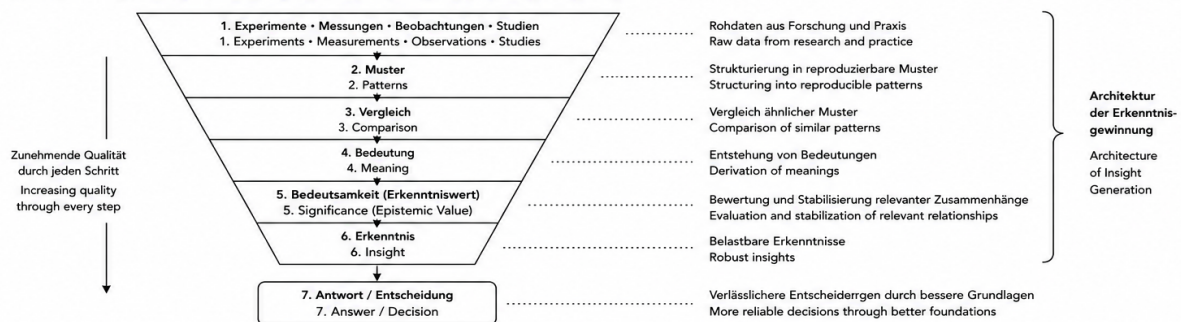
GRAFIK 2A: VON DATEN ZU ERKENNTNIS – DIE ARCHITEKTUR DES VERGLEICHS

FIGURE 2A: FROM DATA TO INSIGHT – THE ARCHITECTURE OF COMPARISON



GRAFIK 2B: FORSCHUNGSSTRICHTER – VON VIELEN DATEN ZU BELASTBARER ERKENNTNIS

FIGURE 2B: RESEARCH FUNNEL – FROM MANY DATA POINTS TO ROBUST INSIGHT



from Facts of Pattern to Insights

Closing Reflection

The future of artificial intelligence will not be determined by larger models alone.

It will be determined by the quality of the knowledge foundations upon which those models operate.

A scientific insight does not become reliable because it appears statistically probable.

It becomes reliable because it emerges from reproducible, comparable, and condensed spaces of experience.

Only what can be compared can be condensed.

Only what can be condensed can generate meaning.

Only what generates meaning can become insight.

The next step in the evolution of artificial systems is therefore not to replace probability.

It is to improve the foundations upon which probability itself depends.

This is where the transition from the theory of cognition to the theory of knowledge architecture begins.